CHAPTER B

Automated banking

Two questions per meeting, in 6 to 8 minutes correction included.
They do not constitute a list to be completed in full at each course.
This is based on the expected teaching, to be shifted if the concept has not yet been studied.

R001 · Fractions

Calculate $\frac{2}{3}+\frac{1}{4}$. **Corrected.** $\frac{11}{12}$: common denominator 12.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R002 · Priorities

Calculate $5-2(3-7)$. **Corrected.** 13: parentheses, product, difference.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R003 · Powers

Simplify $\frac{10^{5}\times 10^{-3}}{10^{4}}$. **Corrected.** $10^{-2}=0{,}01$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R004 · Roots

Simplify $\sqrt{50}$. **Corrected.** $5\sqrt{2}$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R005 · Proportion

18 represents 30% of a total. What total? **Corrected.** $\frac{18}{0{,}3}=60$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R006 · Direct rate

A price of 80 drops by 15%. **Corrected.** $80\times 0{,}85=68$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R007 · Initial value

After an increase of 20%, the value is 72. **Corrected.** $\dfrac{72}{1{,}2} = 60$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R008 · Calculated rate

A quantity goes from 40 to 50. **Corrected.** $\dfrac{50-40}{40} = 25\,\%$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R009 · Successive rates

20% increase then 20% decrease: overall rate? **Corrected.** $1{,}2 \times 0{,}8 = 0{,}96$: decrease of 4 per cent.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R010 · Converse rate

What rate compensates for a 20% decrease? **Corrected.** $\dfrac{1}{0{,}8} = 1{,}25$: 25 per cent increase.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R011 · Zero product

Solve (2 $x-1)(x+3)=0$. **Corrected.** $x = \dfrac{1}{2}$ or $x=-3$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R012 · Development

Development $(2x-3)^2$. **Corrected.** $4x^2-12x+9$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R013 · Factorization

Factor 3 x^2-12x. **Corrected.** $3x(x-4)$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R014 · Affine sign

Solve $-3x + 6 \geq 0$. **Corrected.** $x \leq 2$. To reverse inequality to division by -3.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R015 · Slot

Slot per (1; 4) and (3; 0) ? **Corrected.** -2.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R016 · Right

Slope Equation -2 par (1; 4) ? **Corrected.** $y = -2x + 6$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R017 · Weighted average

The values 2 and 5 are effective 3 and 2. **Corrected.** $\frac{6 + 10}{5} = 3{,}2$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R018 · Median

Median of 1, 2, 2, 5, 8, 10? **Corrected.** $\frac{2 + 5}{2} = 3{,}5$ according to the average agreement of the two values power stations.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R019 · Quartiles

For series 1, 2, 3, 4, 5, 6, 7, 8, give Q_1 and Q_3 with rows ceilings $\left\lceil \frac{n}{4} \right\rceil$ and $\left\lceil \frac{3n}{4} \right\rceil$. **Corrected.** $Q_1 = 2$, $Q_3 = 6$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R020 · Statistical standard deviation

For 1 and 3 equiponed, mean, variance and standard deviation? **Corrected.** Average 2, variance $\frac{[(1-2)^2+(3-2)^2]}{2}$ = 1, standard deviation 1.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R021 · Units

Convert 1 min 40 s to seconds; 2400 images over this duration give how much per second? **Corrected.** 100 s; 24 images/s.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R022 · Simple probability

On a balanced dice, probability of a multiple of 3 ? **Corrected.** $\frac{2}{6}=\frac{1}{3}$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R023 · Complementary

$P(A)=\frac{3}{8}$. Provide $P(\overline{A})$. **Corrected.** $\frac{5}{8}$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R024 · Interval

Write $x \geq -2$ and $x <$ Three in the interval. **Corrected.** $[-2;3[$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R025 · Logic

Denying "all the students have succeeded." **Corrected.** "At least one student failed."

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R026 · Histogram

Two classes of widths 2 and 4 have the same number 20. In a histogram with proportional areas what relative heights? **Corrected.** Heights 10 and 5: the area, not the height alone, Number of observations.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R027 · Circular diagram

A sector measures 72 °. Proportion? **Corrected.** $\frac{72}{360} = 20\,\%$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R028 · Scale

An axis starts at 90. Bars stop at 100 and 110. Is the second value double the value of the first? **Corrected.** No: $\frac{110}{100} = 1{,}1$. The heights drawn above 90 deceive this comparison.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R029 · Percentage Percentage

40% of students make A; 25% make B. Proportion that makes A and B? **Corrected.** $0{,}4 \times 0{,}25 = 10\,\%$.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R030 · Proportion and strength

One category represents 0.125 of 80 people. **Corrected.** 10 people.

Prerequisite: Previous Acquis

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R031 · Defining and Representing a Sequence

Calculate the first four terms of $w_0 = 2,\ w_{n+1} = w_n + 2n + 1$; propose a formula explicit. **Corrected.** $2, 3, 6, 11$. The formula $w_n = n^2 + 2$ fits: it gives the first term and $(n+1)^2 + 2 - (n^2 + 2) = 2n + 1$.

Prerequisite: Literal calculation, integer powers, percentages, affine function, Python loop.

Available from 2026-09-09; recommended covers: 2026-09-17, 2026-10-01, 2026-11-03.

R032 · Arithmetic Sequences

We know. $u_3 = 17$ and $u_8 = 37$ for an arithmetic sequence. r and u_0. **Corrected.** $37 - 17 = 5r$, so $r = 4$ then $u_0 = 17 - 3 \times 4 = 5$.

Prerequisite: Literal calculation, integer powers, percentages, affine function, Python loop.

Available from 2026-09-11; recommended covers: 2026-09-22, 2026-10-06, 2026-11-03.

R033 · Geometric sequences

Studying the variation in $u_n = -2 \times 3^n$and then $v_n = 2(-1)^n$. **Corrected.** $u_{n+1} - u_n = -4 \times 3^n < 0$: Decreasing. v_n alternate 2 and -2: neither increasing nor decreasing.

Prerequisite: Literal calculation, integer powers, percentages, affine function, Python loop.

Available from 2026-09-16; recommended repeats: 2026-09-24, 2026-10-08, 2026-11-05.

R034 · Amounts, indices and demonstrations

Calculate $\sum_{k=2}^{5} 2^k$ without summation symbol in the answer. **Corrected.** $4 + 8 + 16 + 32 = 60$, or $\dfrac{4(1-2^4)}{1-2} = 60$.

Prerequisite: Literal calculation, integer powers, percentages, affine function, Python loop.

Available from 2026-09-18; recommended covers: 2026-09-29, 2026-10-13, 2026-11-10.

R035 · Conjecturing a limit and programming a threshold

For $u_n = 48(\frac{1}{-})^n$, determine the first rows where $u_n \leq 6$ and where $u_n < 6$. **Corrected.** $u_3 = 6$: level 3 for ≤ 6, row 4 for < 6.

Prerequisite: Literal calculation, integer powers, percentages, affine function, Python loop.

Available from 2026-09-18; recommended covers: 2026-09-29, 2026-10-13, 2026-11-10.

R036 · Three forms, three readings

Give three forms of $g(x) = x^2 + 2x - 3$. Read its roots and minimum. **Corrected.** $g(x) = (x+1)^2 - 4 = (x-1)(x+3)$. Roots 1 and -3, minimum -4 in -1.

Prerequisite: Remarkable identities, zero product, product sign, square roots.

Available from 2026-09-23; recommended repeats: 2026-10-01, 2026-10-15, 2026-11-12.

R037 · Discrimination and Resolution

Solve 3 $x^2 - 6x + 3 = 0$ and $x^2 + 2x + 5 = 0$. **Corrected.** The first is 3($x - 1)^2 = 0$, therefore 1. For the second one, $-16 < 0$: no real solution.

Prerequisite: Remarkable identities, zero product, product sign, square roots.

Available from 2026-09-25; recommended repeats: 2026-10-06, 2026-11-03, 2026-11-17.

R038 · Factoring without automating the discriminant

Determine the level 2 f polynomial such that $f(1) = f(4) = 0$ and $f(0) = -8$. **Corrected.** $f(x) = a(x-1)(x-4)$; $4a = -8$, so $f(x) = -2(x-1)(x-4)$.

Prerequisite: Remarkable identities, zero product, product sign, square roots.

Available from 2026-09-30; recommended covers: 2026-10-08, 2026-11-03, 2026-11-19.

R039 · Sign, inequities and optimization

Solve $x^2 - 4x - 5 < 0$ then maximize $x(8 - x)$ on [0; 3]. **Corrected.** $(x - 5)(x + 1) < 0$ given $]-1; 5[$. The summit of $x(8-x)$ is in 4, out of the domain ; on [0; 3], the function grows and its maximum is worth 15 in 3.

Prerequisite: Remarkable identities, zero product, product sign, square roots.

Available from 2026-10-02; recommended repeats: 2026-10-13, 2026-11-03, 2026-11-24.

R040 · Projection and angle

Two vectors have norms 2 and 5 and form 90 °, then 180 °. Calculate the products. **Corrected.** 0 then −10. Cosine is equal to 0 and −1.

Prerequisite: Vectors, Chasles· relation, Pythagoras, cosine in a right triangle, orthonormal coordinates.

Available from 2026-10-07; recommended repeats: 2026-10-15, 2026-11-03, 2026-11-26.

R041 · Coordinates, Norm and Orthogonality

For what real t the vectors (t; (2) and (3); −6) Are they orthogonal? **Corrected.** $3t - 12 = 0$, therefore $t = 4$.

Prerequisite: Vectors, Chasles· relation, Pythagoras, cosine in a right triangle, orthonormal coordinates.

Available from 2026-10-09; recommended covers: 2026-11-03, 2026-11-03, 2026-12-01.

R042 · Vector Calculation and Identities

We give $\|\vec{u}\| = 2$, $\|\vec{v}\| = 3$, $\vec{u} \cdot \vec{v} = 1$. Calculate $\|\vec{u} - \vec{v}\|$. **Corrected.** The square is worth $4 - 2 + 9 = 11$; the norm is $\sqrt{11}$.

Prerequisite: Vectors, Chasles· relation, Pythagoras, cosine in a right triangle, orthonormal coordinates.

Available from 2026-10-14; recommended covers: 2026-11-03, 2026-11-05, 2026-12-03.

R043 · Al-Kashi and choice of method

A triangle a $AB = AC = 4$ and $BC = 4\sqrt{2}$. Determine angle A. **Corrected.** $32 = 16+16-32\cos A$, so cos $A = 0$ and $A = 90°$.

Prerequisite: Vectors, Chasles· relation, Pythagoras, cosine in a right triangle, orthonormal coordinates.

Available from 2026-10-16; recommended covers: 2026-11-03, 2026-11-10, 2026-12-08.

R044 · The AB diameter circle

For $A(0; 0)$, $B(6; 0)$, determine M such as $\overrightarrow{MA} \cdot \overrightarrow{MB} = 7$. **Corrected.** $MI^2 - 9 = 7$: circle centre (3; 0) and radius 4.

Prerequisite: Vectors, Chasles· relation, Pythagoras, cosine in a right triangle, orthonormal coordinates.

Available from 2026-10-16; recommended covers: 2026-11-03, 2026-11-10, 2026-12-08.

R045 · Rate of change and secants

Calculate the rate of $f(x) = x^2 + 3x$ between 2 and 2 + h. **Corrected.** $f(2+h) - f(2) = 7h + h^2$; rate $7 + h$ for $h \neq 0$.

Prerequisite: Images, slope of a straight line, literal calculation, square root and absolute value.

Available from 2026-11-04; recommended covers: 2026-11-12, 2026-11-26, 2027-01-05.

R046 · Derivative number

For $f(x) = 3x^2 - 2x$, calculate $f\prime(1)$ from the rate. **Corrected.** The rate is 4 + 3 h, so the derived number is 4.

Prerequisite: Images, slope of a straight line, literal calculation, square root and absolute value.

Available from 2026-11-06; recommended covers: 2026-11-17, 2026-12-01, 2027-01-05.

R047 · Tangente and affine approximation

We give $f(3) = 5$ and $f\prime(3) = -2$. Write the tangent and approach $f(3{,}04)$. **Corrected.** $y = 5 - 2(x-3) = -2x + 11$; approximation $5 - 2(0{,}04) = 4{,}92$.

Prerequisite: Images, slope of a straight line, literal calculation, square root and absolute value.

Available from 2026-11-11; recommended repeats: 2026-11-19, 2026-12-03, 2027-01-05.

R048 · When Differentiability Fails

Justify that $g(x) = |x-2|$ is not derivable in 2. Also study the differentiability of $x \mapsto \sqrt{x}$ in 0 per its rate. **Corrected.** The rate in 2 is $\frac{|h|}{h}$, with the two opposite limit values. in 0, the rate $\frac{1}{\sqrt{h}}$ for $h > 0$ exceeds any limit when h approaches 0: no derived number real.

Prerequisite: Images, slope of a straight line, literal calculation, square root and absolute value.

Available from 2026-11-13; recommended covers: 2026-11-24, 2026-12-08, 2027-01-05.

R049 · Conditioner: change denominator

Out of 60 objects, 20 are red; among the red, 5 are damaged. $P(R \cap D)$ and $P_R(D)$. **Corrected.** $\frac{5}{60} = \frac{1}{12}$ and $\frac{5}{20} = \frac{1}{4}$.

Prerequisite: Fragments, proportions, probabilities on a finite universe, intersection and meeting.

Available from 2026-11-18; recommended repeats: 2026-11-26, 2026-12-10, 2027-01-07.

R050 · Trees, partitions and total probabilities

$P(A) = \frac{1}{3}$, $P_A(B) = \frac{3}{4}$, $P_{\overline{A}}(B) = \frac{1}{4}$. Calculate $P(B)$. **Corrected.** $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$.

Prerequisite: Fragments, proportions, probabilities on a finite universe, intersection and meeting.

Available from 2026-11-20; recommended repeats: 2026-12-01, 2026-12-15, 2027-01-12.

R051 · Independence and Incompatibility

We give $P(A) = 0{,}4$, $P(B) = 0{,}5$, $P(A \cap B) = 0{,}18$. Are they independent? **Corrected.** No: $0{,}4 \times 0{,}5 = 0{,}20 \neq 0{,}18$.

Prerequisite: Fragments, proportions, probabilities on a finite universe, intersection and meeting.

Available from 2026-11-25; recommended repeats: 2026-12-03, 2026-12-17, 2027-01-19.

R052 · Successive Tests and Bernoulli

Four independent probability tests $p = \frac{1}{3}$. Calculation of the probability of four failures and then of less a success. **Corrected.** $(\frac{2}{3})^4 = \frac{16}{81}$ then $1 - \frac{16}{81} = \frac{65}{81}$.

Prerequisite: Fragments, proportions, probabilities on a finite universe, intersection and meeting.

Available from 2026-11-27; recommended repeats: 2026-12-08, 2027-01-05, 2027-01-19.

R053 · Simulate and discuss the model

A simulation of 2000 points gives 675 points under $y = x^2$. Which area does it estimate? **Corrected.** $\frac{675}{2000} = 0{,}3375$; this is an estimate, not the exact value.

Prerequisite: Fragments, proportions, probabilities on a finite universe, intersection and meeting.

Available from 2026-11-27; recommended repeats: 2026-12-08, 2027-01-05, 2027-01-19.

R054 · Derivative function and usual functions

Dirty x^5, x^{-2} and $\sqrt{x}$ specifying the areas. **Corrected.** $5x^4$ on $\mathbb{R}$; $-\frac{2}{x^3}$ for $x \neq 0$; $\frac{1}{2\sqrt{x}}$ for $x > 0$.

Prerequisite: Derivative number, algebraic calculation, trinomial sign, intervals.

Available from 2026-12-02; recommended repeats: 2026-12-10, 2027-01-05, 2027-01-21.

R055 · Sum, product, inverse and quotient

Derivative $(x^2+1)(3x-2)$ then $\frac{1}{x^2+1}$. **Corrected.** $2x(3x-2)+3(x^2+1) = 9x^2-4x+3$; $-\frac{2x}{(x^2+1)^2}$, on $\mathbb{R}$.

Prerequisite: Derivative number, algebraic calculation, trinomial sign, intervals.

Available from 2026-12-04; recommended repeats: 2026-12-15, 2027-01-05, 2027-01-26.

R056 · Sign of derivative and extrema

Study $f(x) = x^2 - 4x + 1$ on [0; 3] and give its extrema over this interval. **Corrected.** $f\prime = 2x - 4$, So decrease to 2 then increase. Minimum -3 in 2 ; values at bounds 1 and -2, so Maximum 1 in 0.

Prerequisite: Derivative number, algebraic calculation, trinomial sign, intervals.

Available from 2026-12-09; recommended repeats: 2026-12-17, 2027-01-05, 2027-01-28.

R057 · Parity, Inequalities and Comparison

Studying gender parity $x^4 + x^2$ and $x^3 - x$; Demonstrate $x^2 + 1 \geq 2x$. **Corrected.** First pair, second odd; difference $x^2 + 1 - 2x = (x-1)^2 \geq 0$, equal to 1.

Prerequisite: Derivative number, algebraic calculation, trinomial sign, intervals.

Available from 2026-12-11; recommended repeats: 2027-01-05, 2027-01-05, 2027-02-02.

R058 · Newton Optimization and Method

A perimeter rectangle 12 m wide x. Determine maximum range by bypass. **Corrected.** $A(x) = x(6-x)$ on [0; 6] ; $A\prime = 6 - 2x$. Maximum 9 m ^{2} in 3.

Prerequisite: Derivative number, algebraic calculation, trinomial sign, intervals.

Available from 2026-12-11; recommended repeats: 2027-01-05, 2027-01-05, 2027-02-02.

R059 · Normal Vector and Cartesian Equation

Equation of the right by (2; -1) normal (3; 4). **Corrected.** $3(x-2) + 4(y+1) = 0$, i.e. $3x + 4y - 2 = 0$.

Prerequisite: Vector coordinates, scalar product, linear systems, square completion.
Always orthonormous.

Available from 2027-01-06; recommended repeats: 2027-01-19, 2027-01-28, 2027-03-02.

R060 · Orthogonal project and distance

Project P(4; 1) on $d : x = 2$. **Corrected.** H(2; 1) and distance 2; normal is horizontal.

Prerequisite: Vector coordinates, scalar product, linear systems, square completion.
Always orthonormous.

Available from 2027-01-08; recommended covers: 2027-01-19, 2027-02-02, 2027-03-02.

R061 · Equation of a circle

Recognize $x^2 + y^2 + 2x - 8y + 17 = 0$. **Corrected.** $(x+1)^2 + (y-4)^2 = 0$: the only point ($-1; 4$), Not a strictly positive circle of radius.

Prerequisite: Vector coordinates, scalar product, linear systems, square completion.
Always orthonormous.

Available from 2027-01-13; recommended repeats: 2027-01-21, 2027-02-04, 2027-03-04.

R062 · Solve a configuration

Intersect ($x-1)^2 + (y+2)^2 = 9$ with $y = 1$. **Corrected.** $(x-1)^2 = 0$, so only point (1; 1); the right is tangent to the circle.

Prerequisite: Vector coordinates, scalar product, linear systems, square completion.
Always orthonormous.

Available from 2027-01-15; recommended repeats: 2027-01-26, 2027-02-09, 2027-03-09.

R063 · Variable, Event and Law

Two balanced pieces are launched. X counts batteries. Write his law and $P(X \leq 1)$. **Corrected.** Values 0.1.2; probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. $P(X \leq 1) = \frac{3}{4}$.

Prerequisite: Probabilities, weighted average, squares, trees and lists.

Available from 2027-01-20; recommended repeats: 2027-01-28, 2027-02-11, 2027-03-11.

R064 · Expectation and linearity

Gross gain equals 10 with probability $\frac{1}{5}$ and 0 otherwise. What bet makes the game fair? **Corrected.** $E(G) = \frac{10}{5}$ = 2. For a bet m, $E(G - m) = 2 - m$; Fairness 2.

Prerequisite: Probabilities, weighted average, squares, trees and lists.

Available from 2027-01-22; recommended repeats: 2027-02-02, 2027-03-02, 2027-03-16.

R065 · Variance, standard deviation and König·Huygens

For X values $-1, 1$ equiprobable, calculate expectation, variance and standard deviation. **Corrected.** $E(X) = 0$, $E(X^2)$ = 1, variance 1 and standard deviation 1.

Prerequisite: Probabilities, weighted average, squares, trees and lists.

Available from 2027-01-27; recommended repeats: 2027-02-04, 2027-03-02, 2027-03-18.

R066 · Transform and Compare Earnings

$E(X) = 3$, $V(X)$ = 4. Give the characteristics of $Y = -2X + 5$. **Corrected.** $E(Y) = -1$, $V(Y) = 16$, $\sigma(Y) = 4$.

Prerequisite: Probabilities, weighted average, squares, trees and lists.

Available from 2027-01-29; recommended repeats: 2027-02-09, 2027-03-02, 2027-03-23.

R067 · Samples and mean observed

For $\mu = 5$, σ = 3 and n=36, give the experimental interval requested. **Corrected.** $\frac{2\sigma}{\sqrt{n}} = 1$; interval [4; 6].

Prerequisite: Probabilities, weighted average, squares, trees and lists.

Available from 2027-01-29; recommended repeats: 2027-02-09, 2027-03-02, 2027-03-23.

R068 · Set a new function

What is the equation of tangent to e^x One? **Corrected.** $y = e + e(x - 1) = ex$.

Prerequisite: Power, geometric sequences, bypass and sign.

Available from 2027-02-03; recommended covers: 2027-02-11, 2027-03-02, 2027-03-25.

R069 · Algebraic properties and comparisons

Simplify $\frac{e^{3x}}{e^{x-1}}$; solve $e^{x+2} > e^{2x}$. **Corrected.** e^{2x+1} ; $x + 2 > 2x$, therefore $x < 2$.

Prerequisite: Power, geometric sequences, bypass and sign.

Available from 2027-02-05; recommended repeats: 2027-03-02, 2027-03-02, 2027-03-30.

R070 · Dériver et étude

Dirty $g(t) = 3e^{-2t}$ and study its sense of variation. **Corrected.** $g\prime(t) = -6e^{-2t} < 0$ on $\mathbb{R}$: Strictly decreasing.

Prerequisite: Power, geometric sequences, bypass and sign.

Available from 2027-02-10; recommended repeats: 2027-03-02, 2027-03-04, 2027-04-06.

R071 · From discreet to continuous

For $Q(t) = 5e^{0,3t}$, express $Q(n)$ and $\frac{Q(t+2)}{Q(t)}$. **Corrected.** $u_n = 5(e^{0,3})^n$; the report of two units $e^{0,6}$.

Prerequisite: Power, geometric sequences, bypass and sign.

Available from 2027-02-12; recommended repeats: 2027-03-02, 2027-03-09, 2027-04-06.

R072 · Digitally build and demonstrate inequality

Show that $e^{2x} \geq 1 + 2x$ for all real x. **Corrected.** Applying demonstrated inequality with argument 2 x ; equality only in x=0.

Prerequisite: Power, geometric sequences, bypass and sign.

Available from 2027-02-12; recommended repeats: 2027-03-02, 2027-03-09, 2027-04-06.

R073 · Measuring an angle in radians

Convert 45 °, 120° ; give the radius arc 2 m and angle $\frac{3\pi}{4}$. **Corrected.** $\frac{\pi}{4}, \frac{2\pi}{3}$; length $\frac{3\pi}{2}$ m.

Prerequisite: Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Available from 2027-03-03; recommended repeats: 2027-03-11, 2027-03-25, 2027-04-27.

R074 · Rewinding and image of a real

Place images of $\frac{7\pi}{2}$ and $-\frac{5\pi}{3}$ Bringing them back to a turn. **Corrected.** $\frac{7\pi}{2}$ In the image of $\frac{3\pi}{2}$; $-\frac{5\pi}{3}$ that of $\frac{\pi}{3}$.

Prerequisite: Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Available from 2027-03-05; recommended repeats: 2027-03-16, 2027-03-30, 2027-04-27.

R075 · Sine, cosine and remarkable values

If $\cos x = \frac{3}{5}$ and image of x is in the fourth quadrant, giving $\sin x$. **Corrected.** $\sin^2 x = \frac{16}{25}$; the sine is negative, so $-\frac{4}{5}$.

Prerequisite: Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Available from 2027-03-10; recommended repeats: 2027-03-18, 2027-04-06, 2027-04-29.

R076 · Associated angles and symmetries

Calculate the two coordinates of the images of $\frac{4\pi}{3}$ and $\frac{\pi}{4}$. **Corrected.** $(-\frac{1}{2}; -\frac{\sqrt{3}}{2})$ and $(\frac{\sqrt{2}}{2}; -\frac{\sqrt{2}}{2})$.

Prerequisite: Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Available from 2027-03-12; recommended repeats: 2027-03-23, 2027-04-06, 2027-05-04.

R077 · Approximate pi and reinvest

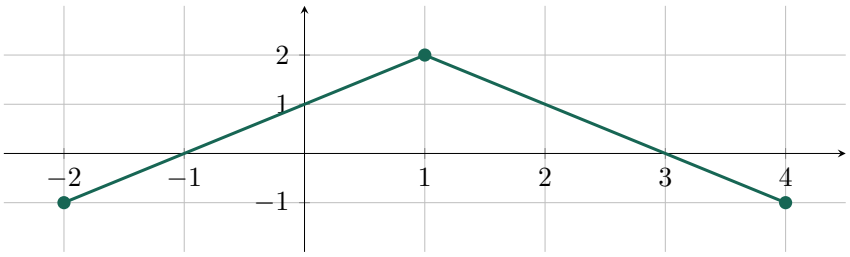
For n=6 and c=1, calculate the new string after doubling. **Corrected.** $h = \frac{\sqrt{3}}{2}$; new rope $\frac{1}{\sqrt{2+\sqrt{3}}} = \sqrt{2-\sqrt{3}}$.

Prerequisite: Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Available from 2027-03-12; recommended repeats: 2027-03-23, 2027-04-06, 2027-05-04.

R078 · Graphical reading

In Figure R078, read $f(1)$, resolve $f(x) = 0$ and $f(x) > 0$.



Corrected. $f(1) = 2$; zeros -1 and 3; positive sign on $]-1;3[$.

Prerequisite: Identification

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.